area of the not-to-exceed zone of the engine, the base emission
control strategy shall ensure that any variation in the point emission
values is progressive and justified from the value at that power on
the line between the mode points, unless rationalized by an auxiliary

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SOURCE: MEPC.397(83).pdf
control strategy shall ensure that any variation in the point emission
values is progressive and justified from the value at that power on
the line between the mode points, unless rationalized by an auxiliary
control device or explained by a physical limitation of the engine;
.5 a declaration that the engine's base emission control strategy only
reacts to changes in engine load and speed;
.6 any other information the applicant considers relevant; and
.7 any other information the Administration requests.
2.5.5 For each auxiliary control device which may operate under steady -state
conditions, the NOx certification pack shall include:
.1 a justification of the need for that device; and
.2 a description for that device, including:
.1 details of the conditions under which that device will
operate and the functioning of that device;
.2 how each modulated parameter of the emission control
system achieves the stated purpose of the base emission
control strategy;
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 7 -

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SOURCE: MEPC.397(83).pdf
control strategy;
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 7 -
.3 the process used to ensure that the modulation is limited to
the conditions where the stated purpose of the auxiliary
control device operational strategy arises and to set the
modulation to be the minimum necessary to achieve that
stated purpose;
.4 the effect of the application of that device on the engine's
base emission control strategy;
.5 for auxiliary control devices that operate above 25% engine
power, the effect on the point emission values shall be
documented;
.6 for auxiliary control devices that operate within the declared
not-to-exceed zone, an estimate of the effect on the point
emission values shall be documented;
.7 any other information the applicant considers relevant; and
.8 any other information the Administration requests.
.3 Auxiliary control devices that only operate during transient
conditions need not be included in the NO x certification pack for
screening.

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SOURCE: MEPC.397(83).pdf
.8 any other information the Administration requests.
.3 Auxiliary control devices that only operate during transient conditions need not be included in the NO x certification pack for screening.
2.5.6 The technical file as required by 2.3.4 shall contain the following information:
.1 identification of those auxiliary control devices declared under 2.5.3;
.2 for those auxiliary control devices covered under 2.5.5, the operating conditions which will cause those devices to function;
.3 the means by which the operation of those auxiliary control devices under 2.5.5 may be verified as part of the onboard NO x verification procedure; and
.4 where the provisions of 2.3.6 apply, the means by which it is to be verified that the required quantities of additional substance used are consistent with achieving the engine's intended base emission control strategy shall be included as part of the onbo ard NO x verification procedure.
2.5.7 Where acceptable to the Administration, the documentation requirements of 2.5.4 and 2.5.5 may alternatively be made by reference to that in respect of marine

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SOURCE: MEPC.397(83).pdf
verification procedure.
2.5.7 Where acceptable to the Administration, the documentation requirements of 2.5.4 and 2.5.5 may alternatively be made by reference to that in respect of marine diesel engines comparable, in terms of NO x emissions characteristics, to the engine to be certified.
2.5.8 The provisions of this section only apply to a marine diesel engine which is installed in a ship as an identical replacement engine if the requirements of this section applied at the time the engine family or engine group to which that engine belongs was first certified."
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 8
Chapter 3 Nitrogen oxides emission standards
3.1 Maximum allowable NOx emission limits for marine diesel engines
4 Paragraph 3.1.4 is replaced by the following:
"3.1.4 In the case of a marine diesel engine to be certified in accordance with paragraph 5.1.1 of regulation 13, the specific emission at each individual mode point shall not exceed the applicable NOx emission limit value by more than 50% except as follows:

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SOURCE: MEPC.397(83).pdf
paragraph 5.1.1 of regulation 13, the specific emission at each individual mode point shall not exceed the applicable NOx emission limit value by more than 50% except as follows:
.1 The 10% mode point in the D2 test cycle specified in 3.2.4.
.2 The 10% mode point in the C1 test cycle specified in 3.2.5.

.3 The idle mode point in the C1 test cycle specified in 3.2.5."
3.2 Test cycles and weighting factors to be applied
5 Section 3.2 is replaced by the following:
"3.2 Test cycles and weighting factors to be applied
3.2.1 For every individual engine or parent engine of an engine family or engine group, one or more of the relevant test cycles specified in 3.2.2 to 3.2.5 shall be applied for verification of compliance with the applicable NOx emission limit contained in regulation 13. Appendix IX provides guidance on the selection of the appropriate test cycle but where discrepancies exist the text of chapter 3 takes precedence.
3.2.2 For a fixed pitch propeller propulsion engine or a propeller -law operated non-propulsion engine, test cycle E3 shall be applied in accordance with table 1.
3.2.3 For a propulsion engine that does not operate with a fixed pitch propeller,

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3.2.2 For a fixed pitch propeller propulsion engine or a propeller -law operated non-propulsion engine, test cycle E3 shall be applied in accordance with table 1.
3.2.3 For a propulsion engine that does not operate with a fixed pitch propeller, including an engine fitted as part of a diesel-electric installation or an engine operated with a controllable-pitch propeller, test cycle E2 shall be applied in accordance with table 2.
3.2.4 For a non-propulsion engine that is a constant -speed engine, test cycle D2 shall be applied in accordance with table 3.
3.2.5 For a non-propulsion engine that operates as a variable -speed engine, not included above, test cycle C1 shall be applied in accordance with table 4.
Table 1 Test cycle for a marine diesel engine meeting paragraph 3.2.2
Test cycle
E3
Speed 100% 91% 80% 63%
Power 100% 75% 50% 25%
Weighting factor 0.2 0.5 0.15 0.15
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 9 -
Table 2 Test cycle for a marine diesel engine meeting paragraph 3.2.3
Test cycle
E2

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AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 9 -
Table 2 Test cycle for a marine diesel engine meeting paragraph 3.2.3
Test cycle
E2
Speed 100% 100% 100% 100%*
Power 100% 75% 50% 25%
Weighting factor 0.2 0.5 0.15 0.15

_____

* There are exceptional cases, including large bore engines intended for E2 applications, in which, owing to their oscillating masses and construction, engines cannot be run at low load at nominal speed without the risk of damaging essential components. In such cases, the engine

manufacturer should make an application to the Administration that the test cycle as given in table 2 above may be modified for the 25% power mode with regard to the engine speed. The adjusted engine speed at 25% power, however, should be as close as possible to the rated engine speed, as recommended by the engine manufacturer and approved by the Administration. The applicable weighting factors for the test cycle should remain unchanged.

Table 3 Test cycle for a marine diesel engine meeting paragraph 3.2.4

Test cycle					
D2					
Speed	100%	100%	100%	100%	100%

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The applicable weighting factors for the test cycle should remain unchanged.

Table 3 Test cycle for a marine diesel engine meeting paragraph 3.2.4

Test cycle					
D2					
Speed	100%	100%	100%	100%	100%
Power	100%	75%	50%	25%	10%
Weighting factor	0.05	0.25	0.3	0.3	0.1

Table 4 Test cycle for a marine diesel engine meeting paragraph 3.2.5

Test					